Chapter 1
Matrices and Systems of Equations

1.3 Matrix Arithmetic 矩阵运算
1.4 Matrix Algebra 矩阵代数

Objectives

- Matrix arithmetic
  - =, +, -, scalar multiplication
  - Matrix-vector multiplication, matrix-matrix multiplication
- Matrix algebra
  - Algebraic properties of matrix operations
  - Transpose, symmetry
  - Inverse, singular/nonsingular matrix

Commutative law

associative law

distributive law.

Notation

- Class and lectures (default by most books)
  - Lower bold face letter as column vector, $\boldsymbol{x} \in R^{n\times 1}$, and its transpose $\boldsymbol{x}^T$ as a row vector.
- Textbook
  - Lower bold face letter as a column vector:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

  - Bold face with arrow as a row vector:

$$\vec{\boldsymbol{y}} = [y_1, y_2, \ldots y_n]$$

transpose.

$a_{ij} \xrightarrow{\text{transpose}} a_{ji}$

Eg. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ 2×3 $m \times n$

$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ 3×2 $n \times m$

$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in R^3$

3×1

$x^T = [1 \quad 2 \quad 3]$

1×3

$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = [1 \quad 2 \quad 3]^T$

$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in R^3$

$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in R^3$

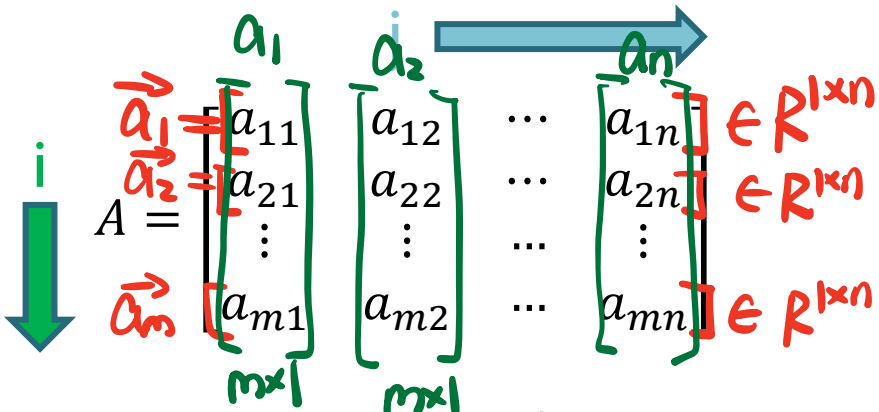
$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \in R^5$

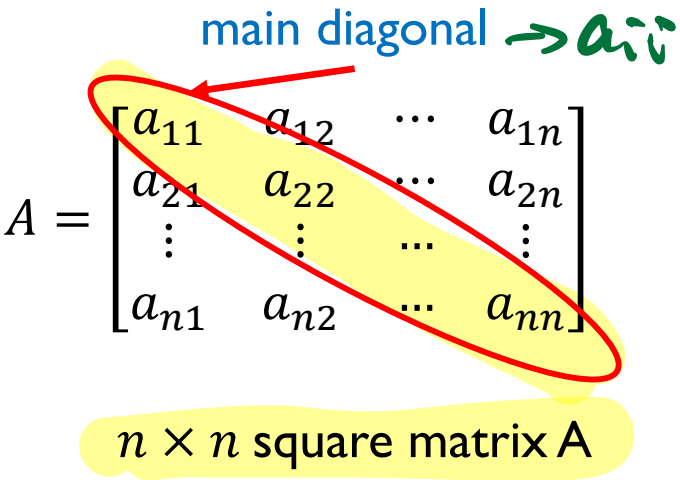
Notation 2

- $\boldsymbol{e}_j = [0, \dots, 0, 1, 0, \dots, 0]^T$ is the column vector whose j-th entry is 1, and other entries are 0.
- For a matrix A,
  - $A = [\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_n]$, $\boldsymbol{a}_j$ is the j-th column.
  - $A = \begin{bmatrix} \vec{\boldsymbol{a}}_1 \\ \vec{\boldsymbol{a}}_2 \\ \vdots \\ \vec{\boldsymbol{a}}_m \end{bmatrix}$, $\vec{\boldsymbol{a}}_i$ is the i-th row.

Matrix arithmetic: Equality

a_{ij}: the entry of matrix A in row i and column j.





$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $A = B$

Definition

Two matrices are **equal** if they are of the same size and if their corresponding entries are all equal.

$a_{ij} = b_{ij}$

Thus, $A = B$ if $a_{ij} = b_{ij} \quad \forall i, j.$ (∀ denotes for every, for all)

Example

If $\begin{bmatrix} a+2b & 2a-b \\ 2c+d & c-2d \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 4 & -3 \end{bmatrix}$, find a, b, c, d.

2×2 2×2

From the matrix, we have:

a+2b = 4 , 2a-b = -2, 2c+d = 4 and c-2d = -3

4×4

$$\begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 2 & -1 & 0 & 0 & -2 \\ 0 & 0 & 2 & 1 & 4 \\ 0 & 0 & 1 & -2 & -3 \end{bmatrix} \xrightarrow[R3\leftrightarrow R4]{R2-2R1} \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & -5 & 0 & 0 & -10 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 2 & 1 & 4 \end{bmatrix} \xrightarrow[R4-2R3]{R2/-5} \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 5 & 10 \end{bmatrix}$$

$$\xrightarrow{R4/5} \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow[R3+2R4]{R1-2R2} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

R.R.E. F. (Reduced Row Echelon Form)

So we have: a = 0, b = 2, c = 1, d = 2

R.E. F. (Row Echelon Form)

Matrices arithmetic: Addition

Definition

Let A and B be matrices of the same size.

Their **sum** $A + B$ is the matrix obtained by adding together the corresponding entries of A and B.

The matrix $A + B$ will be of the same size as A and B.

If A and B are not of the **same size**, they cannot be added, and we say that **the sum does not exist**.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}$$

Thus if $C = A + B$, then $c_{ij} = a_{ij} + b_{ij} \quad \forall i, j.$

Example

Let $\mathbf{A} = \begin{bmatrix} 1 & 4 & 7 \\ 0 & -2 & 3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 2 & 5 & -6 \\ -3 & 1 & 8 \end{bmatrix}$, and $\mathbf{C} = \begin{bmatrix} -5 & 4 \\ 2 & 7 \end{bmatrix}$.

2×3 2×3 2×2

Determine $A + B$ and $A + C$, if the sum exist.

2×3 2×2

doesn't exist

$$A+B=\begin{bmatrix} 3 & 9 & 1 \\ -3 & -1 & 11 \end{bmatrix}$$

3

Scalar multiplication of matrices

1×1 ∈ R

Definition

Let A be a matrix and c be a scalar. The **scalar multiple** of A by c, denoted cA, is the matrix obtained by multiplying every entry of A by c. The matrix cA will be the same size as A.

A: m×n

$cA = c[a_{ij}] = [ca_{ij}]$

$cA \in \mathbb{R}^{m \times n}$

Example

$$\text{Let } A = \begin{bmatrix} 1 & -2 & 4 \\ 7 & -3 & 0 \end{bmatrix}. \quad 2 \times 3$$

$$3A = \begin{bmatrix} 3 & -6 & 12 \\ 21 & -9 & 0 \end{bmatrix}$$

Negation and Subtraction

Definition

- The matrix (-1)*C* is written –*C* and is called the **negative** of *C*.
- We define **subtraction** in terms of addition and scalar multiplication.

$$A - B = A + (-1)B$$

(1) same size (2) $C = A - B$

$c_{ij} = a_{ij} - b_{ij}$

Example

$$C = \begin{bmatrix} 1 & 0 & -7 \\ -3 & 6 & 2 \end{bmatrix}, \text{ then } -C = \begin{bmatrix} -1 & 0 & 7 \\ 3 & -6 & -2 \end{bmatrix}$$

$$\text{Suppose } A = \begin{bmatrix} 5 & 0 & -2 \\ 3 & 6 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 8 & -1 \\ 0 & 4 & 6 \end{bmatrix}.$$

2×3 2×3

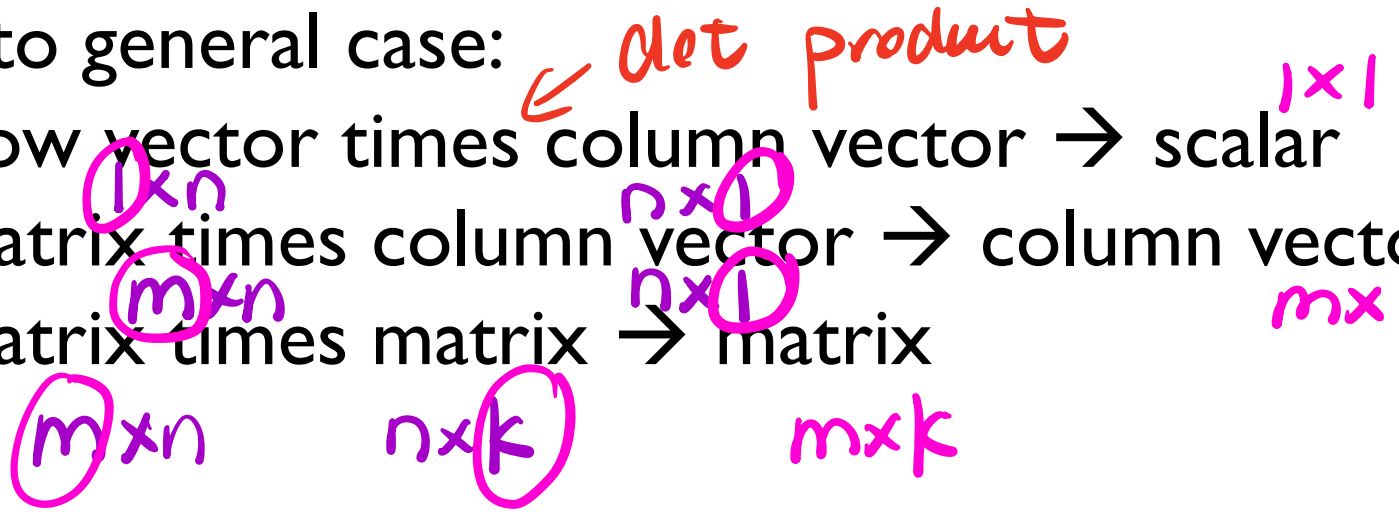
$$A - B = \begin{bmatrix} 5-2 & 0-8 & -2-(-1) \\ 3-0 & 6-4 & -5-6 \end{bmatrix} = \begin{bmatrix} 3 & -8 & -1 \\ 3 & 2 & -11 \end{bmatrix}.$$

Matrix multiplication

Given an mxn matrix A and an n-vector x, we define their product Ax (as an m-vector) so that the system of linear equation can be written as Ax = b.

In the following slides we will see this from simple case to general case:

- Row vector times column vector → scalar
- Matrix times column vector → column vector
- Matrix times matrix → matrix



Product between row and column vectors

(sometimes called the dot product, or inner product, of 2 vectors)

Given a row vector and a column vector,

$$\mathbf{a}^T = [a_1 \;\; a_2 \;\cdots\; a_n\,], \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix},$$

We define their product as a number,

$$\mathbf{a}^T\boldsymbol{x} = a_1x_1 + a_2x_2 + \cdots\; a_nx_n = \sum_{i=1}^{n} a_ix_i$$

Example 1:

$\mathbf{a}^T\mathbf{b} = 1(4) + 2(-1) = 2$

- If $\mathbf{a}^T = [1 \;\; 2]$, $\mathbf{b} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$, find $\mathbf{a}^T\mathbf{b}$.
- If $\mathbf{x}^T = [5 \;\; 1 \;\; 4]$, $\mathbf{y} = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}$, find $\mathbf{x}^T\mathbf{y}$.

$\mathbf{x}^T\mathbf{y} = 5(3) + 1(4) + 4(2) = 27$

  - Meaning: total price for 3 items.



Q: Given $x \in \mathbb{R}^n$

(1) Can $x^T x \geq 0$ be negative? No.

(2) If $x^T x = 0$, what is x?

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

$$x^T x = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= x_1^2 + x_2^2 + \cdots + x_n^2 \geq 0$$

$$\geq 0 \quad \geq 0 \qquad \geq 0$$

$$x^T x = x_1^2 + x_2^2 + \cdots + x_n^2 = 0$$

$$= 0 \quad = 0 \qquad = 0$$

$\Rightarrow x_1 = 0$ & $x_2 = 0$ & $\cdots$ & $x_n = 0$

$$x = [0, 0, \cdots, 0]^T \in \mathbb{R}^n$$

Example 2

- The linear equation

$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$

can be written as

$$[a_1, a_2, \ldots, a_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = b$$

$x \in \mathbb{R}^n$

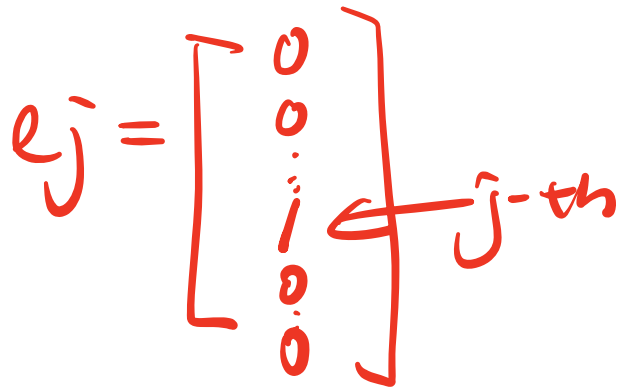
$$A\boldsymbol{x} = b$$

Where $A = [a_1, a_2, \ldots, a_n]$ is a $1 \times n$ matrix.

Example 3

- Suppose

$$e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$



Then

$$\boldsymbol{e}_3{}^T\boldsymbol{e}_2 = 0,$$
$$\boldsymbol{e}_2^T\boldsymbol{e}_2 = 1 = \boldsymbol{e}_3^T\boldsymbol{e}_3$$

In general, the product

$$\boldsymbol{e}_i^T\boldsymbol{e}_j = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Kronecker function

Example before matrix-vector multiplication

- Knowing the price and volume of each drink, and knowing how many drinks there are, calculate the total price and total volume.

	coconut water	cane juice	milk
price (¥)	5	1	4
volume (ml)	330	250	200

	quantity
coconut water	3
cane juice	4
milk	2

- Total price is the product:

$$\begin{bmatrix}5 & 1 & 4\end{bmatrix}\begin{bmatrix}3\\4\\2\end{bmatrix} = 5\times3 + 1\times4 + 4\times2 = 27$$

$1\times n$ $\quad n\times 1$ $\quad 1\times 1$

- Total volume is also a product:

$$\begin{bmatrix}330 & 250 & 200\end{bmatrix}\begin{bmatrix}3\\4\\2\end{bmatrix} = 330\times3 + 250\times4 + 200\times2 = 2390$$

- Matrix-vector multiplication is doing 2 products at the same time:

$$\begin{bmatrix}5 & 1 & 4\\330 & 250 & 200\end{bmatrix}\begin{bmatrix}3\\4\\2\end{bmatrix} = \begin{bmatrix}27\\2390\end{bmatrix}$$

$\vec{a}_1$ → price → $\vec{a}_1 x$

$\vec{a}_2$ → volume → $\vec{a}_2 x$

$m\times n$ $\quad n\times 1$ $\quad m\times 1$

Matrix-vector multiplication

If A is the m x n matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \vec{a}_1 \\ \vec{a}_2 \\ \vdots \\ \vec{a}_m \end{bmatrix}$$

$m \times n$

and x is the n-vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

$n \times 1$

Then we define the matrix-vector product A**x** as the m-vector

$$Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}$$

$m \times 1$

$$Ax = \begin{bmatrix} \vec{a}_1 x \\ \vec{a}_2 x \\ \vdots \\ \vec{a}_m x \end{bmatrix}$$

$m \times 1$

In other words, if $\vec{a}_i$ is the i-th row of A, then

$$Ax = \begin{bmatrix} \vec{a}_1 x \\ \vec{a}_2 x \\ \vdots \\ \vec{a}_m x \end{bmatrix}$$

Also, if $\boldsymbol{a}_j$ is the j-th column of A, then

$$Ax = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\boldsymbol{a}_n$$

$$Ax = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ a_{21}x_1 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix}_{m\times 1}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{matrix} \to \in \mathbb{R} \\ \to \in \mathbb{R} \\ \\ \to \in \mathbb{R} \end{matrix}$$

$$= \begin{bmatrix} a_{11}x_1 \\ a_{21}x_1 \\ \vdots \\ a_{m1}x_1 \end{bmatrix} + \cdots + \begin{bmatrix} a_{1n}x_n \\ a_{2n}x_n \\ \vdots \\ a_{mn}x_n \end{bmatrix}$$

$m\times 1$ $\qquad$ $m\times 1$

$$= x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

$$= x_1 a_1 + \cdots + x_n a_n$$

x_1: scalar, 1×1 $\qquad$ $a_1 \in \mathbb{R}^m$

x_n: scalar, 1×1 $\qquad$ $a_n \in \mathbb{R}^m$

$$Ax = \begin{bmatrix} \vec{a_1}x \\ \vec{a_2}x \\ \vdots \\ \vec{a_m}x \end{bmatrix} = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n \quad \star\star$$

a linear combination of the columns of A

☆☆☆ Linear combination of vectors.

If $a_1, a_2 \cdots a_n \in \mathbb{R}^m$ and $c_1, c_2, \cdots c_n \in \mathbb{R}$ (scalars)

then $c_1 a_1 + c_2 a_2 + \cdots + c_n a_n$

is a linear combination of vectors $a_1 \cdots a_n$.

scalar multiplication, addition

Eg. $a_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $a_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$3a_1 + 2a_2$

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \\ b_m \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$$

$A\underline{x} = \underline{b}$

A: $m \times n$, coefficient matrix; $\underline{x}$: $n \times 1$; $\underline{b}$: $m \times 1$

Eg. $x+y=3$

$x-y=1$

$\longrightarrow$ $(2,1)^T$ or $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is a solution to this LS.

2×2

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

2×1

$$A \quad x \quad = \quad b$$

Linear system can be represented as a matrix-vector equation

$$A\mathbf{x} = \mathbf{b}$$

$Ax = b$

$$Ax = \begin{bmatrix} \vec{a_1}\, x \\ \vec{a_2}\, x \\ \vdots \\ \vec{a_m}\, x \end{bmatrix}$$

E.g,

$$\begin{aligned} 2x_1 - 3x_2 - 4x_3 &= 15 \\ 2x_1 + 4x_2 + 6x_3 &= -12 \\ 3x_1 + 4x_2 + 5x_3 &= -8 \end{aligned}$$

$$= x_1 a_1 + \cdots + x_n a_n$$

$$A = \begin{bmatrix} 2 & -3 & -4 \\ 2 & 4 & 6 \\ 3 & 4 & 5 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \mathrm{b} = \begin{bmatrix} 15 \\ -12 \\ -8 \end{bmatrix}$$

As a matrix equation: $\begin{bmatrix} 2 & -3 & -4 \\ 2 & 4 & 6 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 15 \\ -12 \\ -8 \end{bmatrix}$

a_1 a_2 a_3 b

As a vector equation: $x_1 \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 4 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ -12 \\ -8 \end{bmatrix}$

☆☆☆ Consistency theorem for linear systems ($\forall$ $m \times n$)

(a) A linear system $Ax=b$ is consistent

if and only if ↓↑

(b) b can be written as a linear combination of column vectors of $A_{m \times n}$

$a_1, a_2, \cdots, a_n$

$\exists c_1, c_2, \cdots, c_n \in R$, such that $b = c_1a_1 + c_2a_2 + \cdots + c_na_n$

(a) → (b) Since $Ax=b$ is consistent

the LS has a solution.

Suppose $(s_1, s_2, \cdots, s_n)$ is a solution.

$Ax=b \rightarrow x_1a_1 + x_2a_2 + \cdots + x_na_n = b$

let $x_1=s_1$, $x_2=s_2$ $\cdots$, $x_n=s_n$

$s_1a_1 + s_2a_2 + \cdots + s_na_n = b$

↓

b can be written as a linear combination of $a_1 \cdots a_n$.

(b) → (a). Since $b = c_1a_1 + c_2a_2 + \cdots + c_na_n$

$c_1, c_2, \cdots, c_n$ exist.

$(c_1, c_2, \cdots c_n)^T$ is a solution to $Ax=b$

↓

Consistent

Eg1 A: 3×2 $\quad Ax=b$

$b = 2a_1 + 3a_2$.

↓

$(2,3)^T$ is a solution to the LS.

Eg2 Let A be a 3×4 matrix.

If $b = a_1 + a_2 + a_3 + a_4$

How many solutions to $Ax=b$?

Since $b = 1a_1 + 1a_2 + 1a_3 + 1a_4$

① $(1, 1, 1, 1)^T$ is a solution to LS.

↓

consistent.

② Since $m < n$ $(3 < 4)$ the LS is impossible to have the unique solution

① & ② ⇒ infinitely many solutions.

Eg 3. Let A be a 5×3 matrix.

If $b = a_1 + a_2 = a_2 + a_3$.

How many solutions to $Ax = b$?

$$\begin{cases} b = a_1 + a_2 = 1 \cdot a_1 + 1 \cdot a_2 + 0 \cdot a_3 \to (1, 1, 0)^T \\ b = a_2 + a_3 = 0 \cdot a_1 + 1 \cdot a_2 + 1 \cdot a_3 \to (0, 1, 1)^T \end{cases}$$

2 solutions → infinitely many solutions.

Eg.4. $Ax=0$ $A: m\times 3$

$2a_1+3a_2=-5a_3$

How many solutions ?

$2a_1+3a_2+5a_3=0$

b.

$(2,3,5)^T$ is a non-trivial solution to $Ax=0$

Since $(0,0,0)^T$ is a solution to $Ax=0$

infinitely many.

Eg.5: $A: 2\times 3$ $Ax=0$?

How many solutions to $Ax=0$?

$m<n$

$\Downarrow$

$Ax=0$ has infinitely many

From matrix-vector to matrix-matrix multiplication

- If two linear systems share the same coefficient matrix but with different right-hand sides, say,

$$A\boldsymbol{x} = \boldsymbol{b}_1 \text{ and } A\boldsymbol{y} = \boldsymbol{b}_2,$$

then we can solve the two linear systems at the same time, by writing the augmented matrix together:

$$[A \mid \boldsymbol{b}_1\ \boldsymbol{b}_2]$$

- In matrix equation form:

$$A[\boldsymbol{x}\ \boldsymbol{y}] = [\boldsymbol{b}_1\ \boldsymbol{b}_2]$$

  where A, $[\boldsymbol{x}\ \boldsymbol{y}]$, $[\boldsymbol{b}_1\ \boldsymbol{b}_2]$ are all matrices.
- So, doing several matrix-vector multiplications in parallel can be rewritten as one matrix-matrix multiplication (see next slide for a definition)
- Dot product $\xrightarrow{\text{in parallel}}$ matrix-vector multiplication $\xrightarrow{\text{in parallel}}$ matrix multiplication

$$\begin{cases} x+y=5 \\ x-y=1 \end{cases} \qquad \begin{cases} x+y=2 \\ x-y=0 \end{cases}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 1 & -1 & 1 \end{array}\right] \qquad \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & -1 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 1 & 5 & 2 \\ 1 & -1 & 1 & 0 \end{array}\right]$$

Example: 鸡兔同笼

	鸡	兔
头	1	1
脚	2	4

	小笼	大笼
鸡	2	5
兔	3	4

	小笼	大笼
头	?	?
脚	?	?

$A_{1\times n}$ $n\times 1$ $\quad x_1 \quad x_2 \quad\quad b_1 \quad b_2$

$$Ax = \begin{bmatrix} \vec{a}_1 x \\ \vdots \\ \vec{a}_m x \end{bmatrix}$$

$m\times n$ $n\times 1$

$$A \cdot B = [Ab_1 \quad Ab_2]$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 16 & 26 \end{bmatrix}$$

2×2 $\quad 2\times 2$

$$Ab_1 = \begin{bmatrix} [1 \quad 1]\begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ [2 \quad 4]\begin{bmatrix} 2 \\ 3 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$$C = AB = A\begin{bmatrix} b_1 & b_2 & \cdots & b_k \end{bmatrix}$$

$m \times n \quad n \times k$ $= \begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_k \end{bmatrix}$

$$Ab_j = \begin{bmatrix} \vec{a_1} b_j \\ \vec{a_2} b_j \\ \vdots \\ \vec{a_m} b_j \end{bmatrix} \qquad C = \begin{bmatrix} \vec{a_1} b_1 & \vec{a_1} b_2 & \cdots & \vec{a_1} b_k \\ \vec{a_2} b_1 & \vec{a_2} b_2 & \cdots & \vec{a_2} b_k \\ \vdots & \vdots & \ddots & \vdots \\ \vec{a_m} b_1 & \vec{a_m} b_2 & \cdots & \vec{a_m} b_k \end{bmatrix}$$

$m \times n$

$$c_{ij} = \vec{a_i} b_j$$

① A: $m \times n$ B: $n \times k$ AB

② $c_{ij} = \vec{a_i} b_j$

Matrix Multiplication

Definition

If the number of columns in a matrix A equals the number of rows in a matrix B, then product matrix AB exists.

Let A: $m \times n$ matrix, B: $n \times k$ matrix,

The **product matrix** $C = AB$ is a $m \times k$ matrix, with entries

$$c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

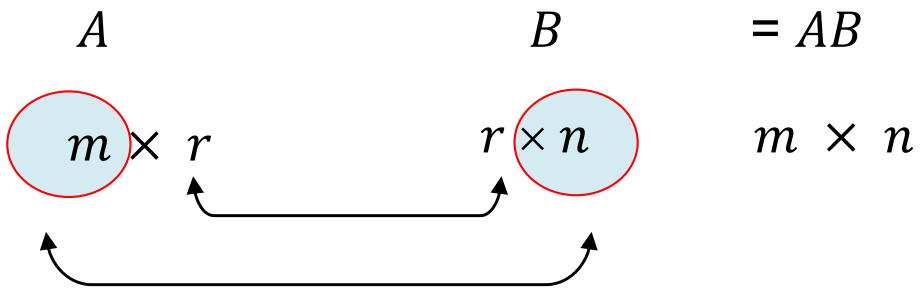
$\vec{a_i}\, b_j$

- The entry c_{ij} is the dot product of the ith row of A, and the jth column of B: $c_{ij} = \vec{\boldsymbol{a}}_{\boldsymbol{i}}\boldsymbol{b}_{\boldsymbol{j}}$
- Also, if $\boldsymbol{b}_j$ is the j-th row of B, then $AB = [A\boldsymbol{b_1}\ A\boldsymbol{b_2}\ A\boldsymbol{b_3}\ \ldots A\boldsymbol{b_k}]$.

If the number of columns in A does not equal the number of row B, **the product does not exist**.

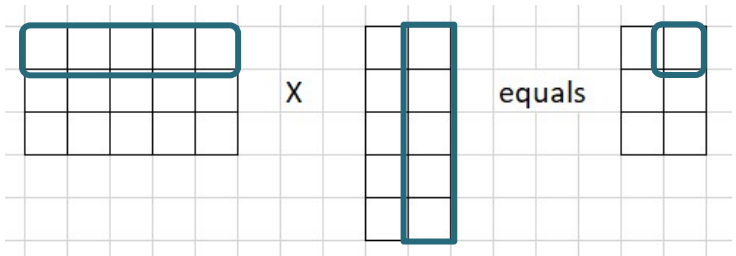
Size of a Product Matrix

If A is an $m \times r$ matrix and B is an $r \times n$ matrix, then AB will be an $m \times n$ matrix.



Example

If A is a 3×5 matrix and B is an 5×2 matrix.
A has five columns and B has five rows, thus AB exits, and AB *is* a 3×2 matrix.



Example

$$c_{ij} = \vec{a}_i \, b_j$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 26 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 1 \quad 2 \times 1$

$$\begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 1 \\ 3 & -2 & 6 \end{bmatrix} = \begin{bmatrix} 14 & -6 & 19 \\ 10 & 0 & 2 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 3 \quad 2 \times 3$

Example

If $A = \begin{bmatrix} -1 & 2 & 5 \\ 0 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -1 & 4 \end{bmatrix}$, find AB and BA.

2×3 3×2

If $A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 5 & -2 \\ -1 & 4 & 0 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$, find Ax.

$c_{ij} = \vec{a_i} b_j$

Solution

$$AB = \begin{bmatrix} -1 & 2 & 5 \\ 0 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} -6 & 13 \\ -3 & 16 \end{bmatrix}$$

c_{11} c_{12} c_{21} c_{22} 2×2

2×3 3×2

$$BA = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 2 & 5 \\ 0 & -2 & 3 \end{bmatrix} = \begin{bmatrix} -1 & -4 & 14 \\ 0 & 4 & -6 \\ 1 & -10 & 7 \end{bmatrix}$$

3×2 2×3 3×3

$AB \neq BA$

$$Ax = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 5 & -2 \\ -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ -9 \end{bmatrix}$$

3×3 3×1 3×1

Matrix multiplication and scalar multiplication

矩阵乘法不满足交换律!

- Scalar: ab=ba, commutative
- Matrix: AB might not equal BA

not commutative

Example 1 矩阵乘法交换律不成立

Let $A = [1\ ,0]$ and $B = \begin{bmatrix}1\\0\end{bmatrix}, C = \begin{bmatrix}0\\1\end{bmatrix},$

Compute AB, BA, AC, CA

Solution $AB = 1;\ AC = 0;\ BA = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, CA = \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}$

Note. In general, $AB \neq BA$.

Example 2 矩阵乘法交换律不成立

Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 1 & 5 \end{bmatrix}, B = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$

AB doesn't exist.

$$BA = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}\begin{bmatrix} 3 & 1 & 2 \\ 4 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} [7 \quad 2]\begin{bmatrix} 3 \\ 4 \end{bmatrix} & [7 \quad 2]\begin{bmatrix} 1 \\ 1 \end{bmatrix} & [7 \quad 2]\begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ [6 \quad 3]\begin{bmatrix} 3 \\ 4 \end{bmatrix} & [6 \quad 3]\begin{bmatrix} 1 \\ 1 \end{bmatrix} & [6 \quad 3]\begin{bmatrix} 2 \\ 5 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 29 & 9 & 24 \\ 30 & 9 & 27 \end{bmatrix}$$

However if we try to calculate AB,

$$AB = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 1 & 5 \end{bmatrix}\begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} [3 \quad 1 \quad 2]\begin{bmatrix} 7 \\ 6 \end{bmatrix} & [3 \quad 1 \quad 2]\begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ [4 \quad 1 \quad 5]\begin{bmatrix} 7 \\ 6 \end{bmatrix} & [4 \quad 1 \quad 5]\begin{bmatrix} 2 \\ 3 \end{bmatrix} \end{bmatrix}$$

can not multiply, therefore AB does not exist.

Note. In general, $AB \neq BA$.

Algebraic Properties of Matrix Operations

Matrix addition and scalar multiplication

Theorem

Let A, B, and C be matrices and a, b, and c be scalars. Assume that the size of the matrices are such that the operations can be performed.

Properties of Matrix Addition and scalar Multiplication

1. $A + B = B + A$ — *Commutative property of addition* 加法交换律
2. $A + (B + C) = (A + B) + C$ — *Associative property of addition* 加法结合律
3. $A + O = O + A = A$ — (*where O is the appropriate zero matrix*)
4. $c(A + B) = cA + cB$ — *Distributive property of scalar multiplication*
5. $(a + b)C = aC + bC$ — *Distributive property of scalar multiplication* 标量乘法分配律
6. $(ab)C = a(bC)$

$2+3=3+2$

1. $A+B=B+A$

$A+B=[a_{ij}]+[b_{ij}]=[a_{ij}+b_{ij}]$

$=[b_{ij}+a_{ij}]=[b_{ij}]+[a_{ij}]=B+A$

2. $A+(B+C)=(A+B)+C$

$A+(B+C)=[a_{ij}]+[b_{ij}+c_{ij}]$

$=[a_{ij}+(b_{ij}+c_{ij})]=[(a_{ij}+b_{ij})+c_{ij}]$

$=[a_{ij}+b_{ij}]+[c_{ij}]=(A+B)+C$

3. $O=\begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m\times n}$

zero matrix

4. $c(A+B)=cA+cB$.

$c(A+B)=c[a_{ij}+b_{ij}]=[c(a_{ij}+b_{ij})]$

$=[ca_{ij}+cb_{ij}]=[ca_{ij}]+[cb_{ij}]=cA+cB$

5. $(a+b)C = aC + bC$

$(a+b)[c_{ij}] = [(a+b)c_{ij}] = [ac_{ij} + bc_{ij}]$

$= [ac_{ij}] + [bc_{ij}] = aC + bC$

6. $(ab)C = a(bC)$

$(ab)C = (ab)[c_{ij}] = [(ab)c_{ij}]$

$= [a(bc_{ij})] = a[bc_{ij}] = a(bC)$

Algebraic Properties of Matrix Operations

Matrix multiplication

Theorem

Let A, B, and C be matrices and a, b, and c be scalars. Assume that the size of the matrices are such that the operations can be performed.

Properties of Matrix Multiplication

AB≠BA.

1. $A(BC) = (AB)C$ — *Associative property of multiplication 乘法结合律*
2. $A(B + C) = AB + AC$ — *Distributive property of multiplication 乘法分配律*
3. $(A + B)C = AC + BC$ ≠C(A+B) — *Distributive property of multiplication 乘法分配律*
4. $AI_n = I_nA = A$ — (*where I_n is the appropriate identity matrix*)
5. $c(AB) = (cA)B = A(cB)$

A:mxn

Note: $AB \neq BA$ in general. *Multiplication of matrices is not commutative.*

不满足交换律

An $n \times n$ identity matrix is the matrix $I_n = (\delta_{ij})$, *where* $\delta_{ij} = \begin{cases} 1 & if\ i = j \\ 0 & if\ i \neq j \end{cases}$

C=AB $c_{ij} = \vec{a_i} b_j$

1. $A(BC) = (AB)C$

$(AB)CD = A(BC)D = AB(CD)$

$(AB)C \neq C(AB)$

$\forall i,j \quad [A(BC)]_{ij} = [(AB)C]_{ij}$

$A: m \times n \quad B: n \times p \quad C: p \times q$

let $D = BC$ $(n \times q)$ $F = AB$ $(m \times p)$

$n \times p \; p \times q$ $m \times n \; n \times p$

$$[A(BC)]_{ij} = [AD]_{ij} = \vec{a_i} d_j = \sum_{k=1}^{n} a_{ik} d_{kj}$$

$1 \times n \quad n \times 1$

$\vec{a_i} = [a_{i1} \; a_{i2} \cdots a_{in}] \qquad d_j = \begin{bmatrix} d_{1j} \\ d_{2j} \\ \vdots \\ d_{nj} \end{bmatrix}$

$$d_{kj} = [BC]_{kj} = \vec{b_k} c_j = \sum_{t=1}^{p} b_{kt} c_{tj}$$

$1 \times p \quad p \times 1$

$$[A(BC)]_{ij} = \sum_{k=1}^{n} a_{ik} \sum_{t=1}^{p} b_{kt} c_{tj}$$

$$= \sum_{k=1}^{n} \sum_{t=1}^{p} a_{ik} b_{kt} c_{tj}$$

$$[(AB)C]_{ij} = [FC]_{ij} = \vec{f}_i \, c_j = \sum_{t=1}^{p} f_{it} c_{tj}$$

$1 \times p \quad p \times 1$

$$f_{it} = [AB]_{it} = \vec{a}_i \, b_t = \sum_{k=1}^{n} a_{ik} b_{kt}$$

$1 \times n \quad n \times 1$

$$[(AB)C]_{ij} = \sum_{t=1}^{p} \left(\sum_{k=1}^{n} a_{ik} b_{kt} \right) c_{tj}$$

$$= \sum_{t=1}^{p} \sum_{k=1}^{n} a_{ik} b_{kt} c_{tj} = [A(BC)]_{ij}$$

$$= \sum_{k=1}^{n} \sum_{t=1}^{p} a_{ik} b_{kt} c_{tj}$$

tg. $2 \sum_{t=1}^{n} t = \sum_{t=1}^{n} 2t$

2. $A(B+C) = AB+AC$

$A(B+C) \neq (B+C)A$

$\neq BA+CA$

$(B+C)A = BA+CA$

$AB+C=D$

$E(AB+C)=ED$

$(AB+C)E=DE$

$E(AB+C)=DE$ X

$[A(B+C)]_{ij} = \vec{a_i}[B+C]_j$

$= \vec{a_i}(b_j + c_j)$

$= \sum_{k=1}^{n} a_{ik}(b_{kj} + c_{kj})$

$= \sum_{k=1}^{n} (a_{ik}b_{kj} + a_{ik}c_{kj})$

$= \sum_{k=1}^{n} a_{ik}b_{kj} + \sum_{k=1}^{n} a_{ik}c_{kj}$

$= \vec{a_i}b_j + \vec{a_i}c_j = [AB]_{ij} + [AC]_{ij}$

$= [AB+AC]_{ij}$

4. $A: m \times n$

$$\underset{m\times n}{A}\underset{n\times n}{I_n} = A \qquad \underset{m\times m}{I_m}\underset{m\times n}{A} = A$$

5. $c(AB) = (cA)B = A(cB)$

$$c(ABCD) = AB(cC)D$$
$$= ABC(cD)$$

$\forall i,j$

$$[c(AB)]_{ij} = c[AB]_{ij} = c\,\vec{a}_i b_j$$

$$= c\sum_{k=1}^{n} a_{ik}b_{kj} = \sum_{k=1}^{n}(ca_{ik})b_{kj} = \vec{[cA]}_i b_j$$

$$\sum_{k=1}^{n} a_{ik}(cb_{kj}) \qquad = [(cA)B]_{ij}$$

$$\vec{a}_i\,[cB]_j = [A(cB)]_{ij}$$

Proof (*A*+*B*=*B*+*A*)

Consider the (*i*,*j*)th elements of matrices *A*+*B* and *B*+*A*:

$$(A+B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B+A)_{ij}.$$

$\therefore$ *A*+*B*=*B*+*A*

Proof A(BC)=(AB)C (optional, a bit messy)

Suppose the size of A, B, C are $m \times n$, $n \times p$ and $p \times q$.

Consider the (*i*,*j*)th elements of matrices A(*BC) and (AB)C*:

$$[A(BC)]_{ij} = \sum_{k=1}^{n} a_{ik}(BC)_{kj} = \sum_{k=1}^{n} a_{ik}\left(\sum_{t=1}^{p} b_{kt}c_{tj}\right)$$

Size of BC is $n \times q$

$$= \sum_{k=1}^{n}\sum_{t=1}^{p} a_{ik}b_{kt}c_{tj}$$

$$[(AB)C]_{ij} = \sum_{t=1}^{p} (AB)_{it}c_{tj} = \sum_{t=1}^{p} \left(\sum_{k=1}^{n} a_{ik}b_{kl}\right)c_{tj}$$

Size of AB is $m \times p$

$$= \sum_{t=1}^{p}\sum_{k=1}^{n} a_{ik}b_{kt}c_{tj} = \sum_{k=1}^{n}\sum_{t=1}^{p} a_{ik}b_{kt}c_{tj}$$

Example

$(AB)C = A(BC)$

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix}$, and $C = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$. Compute ABC.

Solution

A	B	C	=	D
2×2	2×3	3×1		2×1

$ABC = (AB)C = A(BC)$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 3 & 11 \end{bmatrix}.$$

2×2 $\quad 2\times 3$ $\quad 2\times 3$

$$(AB)C = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 3 & 11 \end{bmatrix}\begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -9 \\ 1 \end{bmatrix}.$$

2×3 $\quad 3\times 1$ $\quad 2\times 1$

Exercise: first calculate (BC), then calculate A(BC), check if you get the same answer.

Caution

In algebra we know that the following **cancellation laws** apply.

- If $ab = ac$ and $a \neq 0$ then $b = c$.
- If $pq = 0$ then $p = 0$ or $q = 0$.

However the corresponding results are not true for matrices.

- $AB = AC$ does not imply that $B = C$.
- $PQ = O$ does not imply that $P = O$ or $Q = O$

Example

(1) Consider the matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} -3 & 8 \\ 3 & -2 \end{bmatrix}$.

Observe that $AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$, but $B \neq C$.

(2) Consider the matrices $P = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$, and $Q = \begin{bmatrix} 2 & -6 \\ 1 & -3 \end{bmatrix}$.

Observe that $PQ = O$, but $P \neq O$ and $Q \neq O$.

Transpose of a Matrices

Definition

The **transpose** of a matrix A, denoted A^T, is the matrix whose columns are the rows of the given matrix A.

Thus $(A^T)_{ij} = A_{ji} \quad \forall i, j.$

A: $m \times n$

A^T: $n \times m$

Example

$$A = \begin{bmatrix} 2 & 7 \\ -8 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & -7 \\ 4 & 5 & 6 \end{bmatrix}, \text{and } C = \begin{bmatrix} -1 & 3 & 4 \end{bmatrix}.$$

a_{21} $[A^T]_{12}$

$$A^T = \begin{bmatrix} 2 & -8 \\ 7 & 0 \end{bmatrix} \quad B^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ -7 & 6 \end{bmatrix} \quad C^T = \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}.$$

Theorem: Properties of Transpose

Let A and B be matrices and c be a scalar. Assume that the sizes of the matrices are such that the operations can be performed.

1. $(A + B)^T = A^T + B^T$ *Transpose of a sum*
2. $(cA)^T = cA^T$ *Transpose of a scalar multiple*
3. $(AB)^T = B^T A^T$ *Transpose of a product*
4. $(A^T)^T = A$

Example

If $A = \begin{bmatrix} 1 & 2 & -3 & 0 \\ 6 & 3 & 5 & -2 \end{bmatrix}$, find A^T.

If $A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$, verify that $(AB)^T = B^T A^T$.

$$AB=\begin{bmatrix}-10 & -6\\ -2 & 6\end{bmatrix}_{2\times 2} \qquad (AB)^T=\begin{bmatrix}-10 & -2\\ -6 & 6\end{bmatrix}$$

$$B^TA^T=\underset{2\times 3}{\begin{bmatrix}0 & 5 & 1\\ 2 & 4 & 0\end{bmatrix}}\underset{3\times 2}{\begin{bmatrix}1 & 3\\ -2 & 0\\ 0 & -2\end{bmatrix}}=\underset{2\times 2}{\begin{bmatrix}-10 & -2\\ -6 & 6\end{bmatrix}}$$

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1. $(A+B)^T = A^T + B^T$

$[A^T]_{ij} = a_{ji}$

$[(A+B)^T]_{ij} = [A+B]_{ji}$

$= a_{ji} + b_{ji}$

$= [A^T]_{ij} + [B^T]_{ij} = [A^T + B^T]_{ij}$

2. $(cA)^T = cA^T$

$[(cA)^T]_{ij} = [cA]_{ji} = c a_{ji} = c[A^T]_{ij}$

$= [cA^T]_{ij}$

3. $(AB)^T = B^T A^T$

$\Downarrow$

$(ABCDE)^T = E^T D^T C^T B^T A^T$

$[A^T]_{ij} = a_{ji}$

$[(AB)^T]_{ij} = [AB]_{ji} = \vec{a}_j b_i = \sum_{k=1}^{n} a_{jk} b_{ki}$

$= \sum_{k=1}^{n} [B^T]_{ik} [A^T]_{kj}$

$= \overrightarrow{[B^T]_i} [A^T]_j = [B^T A^T]_{ij}$

$[A^T]_{kj}$ $[B^T]_{ik}$

4. $(A^T)^T = A$

$$[(A^T)^T]_{ij} = [A^T]_{ji} = a_{ij} = [A]_{ij}$$

Eg. $[D(A+B)C - E]^T [ABC]^T$

$$= [(D(A+B)C)^T - E^T][C^T B^T A^T]$$

$$= [C^T(A+B)^T D^T - E^T] C^T B^T A^T$$

$$= (C^T(A^T+B^T)D^T - E^T) C^T B^T A^T$$

$$= [C^T A^T D^T + C^T B^T D^T - E^T] C^T B^T A^T$$

$$= C^T A^T D^T C^T B^T A^T + C^T B^T D^T C^T B^T A^T - E^T C^T B^T A^T$$

(optional proofs)

Proof for 3. $(AB)^T = B^T A^T$

Suppose the size of A and B are $m \times n$ and $n \times p$.
Consider the (i,j)th elements of matrices $(AB)^T$ *and* $B^T A^T$:

$$(AB)^T_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

$$(B^T A^T)_{ij} = \sum_{k=1}^{n} (B^T)_{ik} (A^T)_{kj}$$
$$= \sum_{k=1}^{n} b_{ki} a_{jk}$$

Therefore, $(AB)^T = B^T A^T$

Symmetric Matrix

$A = A^T$

$m \times n$ $\quad$ $n \times m$

$m = n$

Definition

A **symmetric matrix** is a matrix that is equal to its transpose.

$n \times n$ square matrix

$$A = A^T \; i.e., a_{ij} = a_{ji} \;\; \forall i, j$$

$a_{ij} = [A^T]_{ij} = a_{ji}$ ← symmetric w.r.t main diagonal

Example

match

$$\begin{bmatrix} 2 & 5 \\ 5 & -4 \end{bmatrix}$$

2x2

$$\begin{bmatrix} 0 & 1 & -4 \\ 1 & 7 & 8 \\ -4 & 8 & -3 \end{bmatrix}$$

3x3

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 7 & 3 & 9 \\ -2 & 3 & 2 & -3 \\ 4 & 9 & -3 & 6 \end{bmatrix}$$

match

4x4

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}_{n \times n} \rightarrow \text{symmetric}$$

Example

Let A and B be symmetric matrices of the same size. Let C be a linear combination of A and B. Prove that C is symmetric.

$$C = c_1 A + c_2 B$$

$$C^T = (c_1 A + c_2 B)^T = (c_1 A)^T + (c_2 B)^T$$

$$= c_1 A^T + c_2 B^T$$

Since A, B are symmetric matrices

$A^T = A, \quad B^T = B$

$$= c_1 A + c_2 B$$

$$= C$$

$\Rightarrow C$ is symmetric

Example

The matrix $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ has the property that $A^2 = O$

Is it possible for a nonzero symmetric 2×2 matrix to have this property?

No.

Why?

If $A = \begin{bmatrix} 1 & q & r \\ 5+r & -r & q-r \\ -3 & 5 & 6 \end{bmatrix}$, find q, r so that A is symmetric.

q=2 r=-3

$$\begin{cases} q = 5 + r \\ r = -3 \\ q - r = 5 \end{cases} \Rightarrow \begin{cases} q = 2 \\ r = -3 \end{cases}$$

① Assume $A=\begin{bmatrix} a & b \\ b & c \end{bmatrix}_{2\times 2}$

$$A^2=AA=\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} a & b \\ b & c \end{bmatrix}=\begin{bmatrix} a^2+b^2 & ab+bc \\ ab+bc & b^2+c^2 \end{bmatrix}_{2\times 2}$$

$$\begin{cases} a^2+b^2=0 \\ ab+bc=0 \\ b^2+c^2=0 \end{cases} \Rightarrow \begin{cases} a=0,\ b=0 \\ c=0 \end{cases} = 0$$

$$=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}$$

$$A=\begin{bmatrix} a & b \\ b & c \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Impossible.

If and only if

Let p and q be statements.
Suppose that p implies q (if p then q), written $p \Rightarrow q$,
and that also $q \Rightarrow p$, we say that

"p if and only if q" (iff) $\quad p \Leftrightarrow q$
(p is necessary and sufficient for q)

$p \rightarrow q \;\&\; q \rightarrow p$

Example

Let A and B be symmetric matrices of the same size. Prove that the product AB is symmetric if and only if $AB = BA$.

Proof

*We have to show (a) AB is symmetric $\Rightarrow$ $AB = BA$, and the converse, (b) AB is symmetric $\Leftarrow$ $AB = BA$.

($\Rightarrow$) Let AB be symmetric, then

$AB = (AB)^T$ since AB is symmetric

$= B^T A^T$ by Thm (3)

$= BA$ since A and B are symmetric

Therefore, if AB is symmetric => AB=BA

($\Leftarrow$) Let $AB = BA$, then

$(AB)^T = (BA)^T$ since AB=BA

$= A^T B^T$ by Thm (3)

$= AB$ since A and B are symmetric

Therefore, if AB=BA => AB is symmetric.

Example

Let A be a symmetric matrix. Prove that A^2 is symmetric.

Proof

$$(A^2)^T = (AA)^T = (A^T A^T) = AA = A^2$$

$A^T = A$

$(A^2)^T = A^2$

$A \; A$

A^n is symmetric.

$\frac{1}{A} \times \frac{A}{B} \times$

$A^{-1} \neq \frac{1}{A}$

A^{-1} ✓ AB^{-1} ✓

The Inverse of a Matrix

Definition

Let A be an $n \times n$ matrix. If a matrix B can be found such that $AB = BA = I_n$, then A is said to be **invertible** and B is called the **inverse** of A. If such a matrix B does not exist, then A has no inverse (denote $B = A^{-1}$, and $A^{-k} = (A^{-1})^k$).

Example

Prove that the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ has inverse $B = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$

Proof

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$BA = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Thus $AB = BA = I_2$, proving that the matrix A has inverse B.

$A_{n\times n}$ — invertible / non-invertible

A^{-1}

$AB = BA = I_n$

$B = A^{-1}$

nonsingular

Singular.

$Ax = b$

$n\times n$

$(A^{-1}A)x = A^{-1}b$

$n\times 1$

I_n

$x = A^{-1}b$ ($n\times n$)

unique

If A is nonsingular

A^{-1} exists.

unique

→ LS $n\times n$ has the unique solution

$n\times n$ A is nonsingular

⇓ ⇑

$n\times n$ $Ax = b$ has the unique solution

⇓

It can be simplified to S.I.F

Why we need inverse?

- Solve equation:

$$Ax = b$$
$$A^{-1}Ax = A^{-1}b$$
$$Ix = A^{-1}b$$
$$x = A^{-1}b$$

- Nonsingular = Invertible Matrices
  - A square $n \times n$ matrix A is ***nonsingular*** (invertible) if there exists a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$.
  - The matrix A^{-1}, if it exists, is called the inverse of A. If A does not have an inverse (A^{-1} does not exist), then A is ***singular*** (noninvertible).
  - Coefficient matrix is **nonsingular** = linear system is consistent, with **unique solution**.

$AB=BA=I_n \Rightarrow B=A^{-1}$

Properties of Matrix Inverse

Let A and B be invertible matrices and c a nonzero scalar, Then

1. $(A^{-1})^{-1} = A$
2. $(cA)^{-1} = \frac{1}{c}A^{-1}$
3. $(AB)^{-1} = B^{-1}A^{-1}$
4. $(A^n)^{-1} = (A^{-1})^n$
5. $(A^T)^{-1} = (A^{-1})^T$

Proof $(AB)^T = B^T A^T$

1. By definition, $AA^{-1} = A^{-1}A = I$.

2. $(cA)\left(\frac{1}{c}A^{-1}\right) = c\frac{1}{c}AA^{-1} = I = \left(\frac{1}{c}A^{-1}\right)(cA) = \left(\frac{1}{c}c\right)(A^{-1}A)$

3. $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I = (B^{-1}A^{-1})(AB)$

4. $A^n(A^{-1})^n = \underbrace{A \dots A}_{n} \cdot \underbrace{A^{-1} \dots A^{-1}}_{n} = \underbrace{A \dots A}_{(n-1)} \cdot I \cdot \underbrace{A^{-1} \dots A^{-1}}_{(n-1)} = \cdots = I = (A^{-1})^n A^n$

5. $AA^{-1} = I, (AA^{-1})^T = (A^{-1})^T A^T = I$

   $A^{-1}A = I, (A^{-1}A)^T = A^T(A^{-1})^T = I$

$A^{-1}\cdots(A^{-1}(A^{-1}AA)A)A$

1. $(A^{-1})^{-1} = A$

$A^{-1} = B$

$AB = I_n$ & $BA = I_n$

$A^{-1}A = I_n$ & $AA^{-1} = I_n$ $\Rightarrow$ $A = (A^{-1})^{-1}$

A B $\quad$ B A

5. $(A^T)^{-1} = (A^{-1})^T$

A $\quad$ B.

$(AB)^T = B^T A^T$

$A^T(A^{-1})^T = [A^{-1}A]^T = I^T = I$

I

$(A^{-1})^T A^T = [A A^{-1}]^T = I^T = I$

Theorem

The product of nonsingular matrices is still nonsingular.

- If A and B are both nonsingular $n \times n$ matrices, then the product AB is also nonsingular, and $(AB)^{-1} = B^{-1}A^{-1}$.

2 ← 1

- If $A_1, \cdots, A_k$ are all nonsingular $n \times n$ matrices, then the product $A_1A_2 \cdots A_k$ is nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = {A_k}^{-1} \cdots {A_2}^{-1} {A_1}^{-1}$$

Example $AB = BA = I_n$

If $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, verify that

$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$ and $(AB)^{-1} = B^{-1}A^{-1}$

① Since A, B are both nonsingular.

A^{-1} exists. B^{-1} exists.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$$

The inverse of AB exists.

$(AB)^{-1} = B^{-1}A^{-1}$ → AB is nonsingular.

② Since $A_1 \cdots A_k$ are nonsingular.

$A_1^{-1}, \cdots, A_k^{-1}$ exist.

$$(A_1 A_2 \cdots A_{k-1} A_k)(A_k^{-1} A_{k-1}^{-1} \cdots A_2^{-1} A_1^{-1})$$

$$= A_1 A_2 \cdots \underbrace{\left(A_{k-1}\underbrace{(A_k A_k^{-1})}_{I} A_{k-1}^{-1}\right)}_{I} \cdots A_2^{-1} A_1^{-1}$$

$$= I$$

$$(A_k^{-1} \cdots (A_1^{-1})(A_1) \cdots A_k) = I$$

Example

If $A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$, then it can be shown that A^{-1} $= \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$. Use this information to compute $(A^T)^{-1}$.

$$(A^T)^{-1} = (A^{-1})^T = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$$

Solution

$$(A^T)^{-1} = (A^{-1})^T = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}.$$

Theorem

If a matrix has an inverse, then that inverse is unique.

Proof

Let B and C be inverses of A.

Thus $AB = BA = I_n$, and $AC = CA = I_n$.

Multiply both sides of the equation $AB = I_n$ by C.

$$C(AB) = CI_n$$
$$(CA)B = C$$
$$I_nB = C$$
$$B = C$$

Thus an invertible matrix has only one inverse.

[unique, no, at most ...

usually use proof of contradiction].

If A is nonsingular $\rightarrow$ A^{-1} is unique.

Proof of contradiction.

Assume there exists two different matrices B, C. which are both the inverse of A.

$B \neq C$

$B = A^{-1}$ $\Rightarrow$ $AB = BA = I$

$C = A^{-1}$ $\Rightarrow$ $AC = CA = I$

$$B = BI = B(AC) = (BA)C = IC = C$$

conflicts with the assumption $B \neq C$!

The inverse of A is unique.

Theorem

Let $AX = Y$ be a system of n linear equations in n variables.
If A^{-1} exists, the solution is unique and is given by $X = A^{-1}Y$.

Proof

($X = A^{-1}Y$ is a solution.)

Substitute $X = A^{-1}Y$ into the matrix equation.

$$AX = A(A^{-1}Y) = (AA^{-1})Y = I_nY = Y$$

(The solution is unique.)

Let X_1 be any solution, thus $AX_1 = Y$. Multiplying both sides of this equation by A^{-1} gives

$$A^{-1}AX_1 = A^{-1}Y$$
$$I_nX_1 = A^{-1}Y$$
$$X_1 = A^{-1}Y$$

As A^{-1} is unique, therefore the solution is unique.

If $p \rightarrow q$.

If not $q \rightarrow$ not p

If $A_{n \times n}$ is nonsingular $\rightarrow$ $Ax=b$ has the unique solution

If $n \times n$ $Ax=b$ doesn't have the unique solution $\rightarrow$ A is singular.

Conclusions

- Addition, Scalar Multiplication
- Multiplication of Matrices
- Algebraic Properties of Matrix Operations
- Transpose, Symmetry, Inverse